USN. | | | | | | | | | | |

[18MTDS102]

FIRST SEMESTER M.Tech UNIVERSITY EXAMINATION, JANUARY 2019

ADVANCES IN DATABASE MANAGEMENT SYSTEMS

Time: 03 Hours **Maximum Marks:100**

Instructions:

i. Answer any **FIVE** full Questions.

ii. Question **1** and **2** are **compulsory.**

iii. Missing data may be suitably assumed.

1. a. Explain in detail about Database Management System advantages over the file management system. **05**

   b. Explain the role of a database administrator. **05**

   c. Briefly Explain the different data models present. **10**

2. a. Explain the Benefits and features of SQL. **05**

   b. Differentiate OLTP and OLAP. **05**

   c. By considering suitable examples, describe the usage of SQL DDL statements (CREATE TABLE, ALTER TABLE WITH METHODS, DROP TABLE). **10**

3. a. How do we represent null values? Discuss the importance of handling null values. **06**

   b. Consider following schema and write SQL for given statements. **06**

   Client_master(clientno, name, address, city, pincode, state, baldue)

   Product_master(productno, name, profitpercent, unitmeasure, sellprice, costprice)

   Salesman_master(Salesmanno,name,address,city,pincode,state,salary,tgtotget, remarks)

   i. List all the clients who are located in Mumbai.

   ii. Delete all salesmen from salesman_master whose salaries are equal to Rs.3500.

   iii. Destroy the table client_master along with data.

   iv. List the name of all clients having 'a' as the second letter in their names.

   v. Count the number of products having cost price is less than or equal to 500.

   vi. Calculate the average, minimum and maximum sell price of product.

   c. Explain the following: **08**

   i. Key constraints

   ii. Integrity constraints

OR

4 a. For the following relation schema: 04

course (course-id, title, dept_name, credits)

instructor (id, name, dept_name, salary)

section (course-id, sec-id, semester, year, building, room_no, time_slot_id)

teaches (id, course-id, sec-id, semester, year)

i. Find the name of all instructors in the physics department.

ii. Find all the courses taught in the fall 2009 semester but not in Spring semester.

iii. Find the names of all instructors in the Comp. Sci. department together with the course titles of all the courses that the instructors teach.

iv. Find the average salary in each department.

b. Explain following operators with example: 10

i. Arithmetic Operators

ii. Logical Operators

iii. Like Operator

iv. IN Operator

v. BETWEEN Operator

c. Discuss the terms Direct Invocation and Embedded SQL. 06

5. a. Explain about the following clauses with example queries. 06

i. Group by

ii. Order by

iii. Aggregation functions

b. Consider the following Hotel database, primary keys are underlined: 06

hotel (hotel_no, name, address)

room (room_no, hotel_no, type, price)

booking (hotel_no, guest_no, date_from, date_to, room_no)

guest (guest_no, name, address)

Give an expression in SQL for each of the following queries:

i. List the names and addresses of all guests in London, alphabetically ordered by name.

ii. List out hotel name and total number of rooms available.

iii. List the details of all rooms at the Grosvenor Hotel, including the name of the guest staying in the room, if the room is occupied.

iv. List all guests currently staying at the Grosvenor Hotel.

v. List the rooms that are currently unoccupied at the Grosvenor Hotel.

vi. List the number of rooms in each hotel in London.

c. What are subqueries? Give examples for subqueries with the SELECT Statement, insert statement, delete statement and update statement. 08

OR

6. a. Discuss following types of Join with suitable example. **06**

i. Inner Join iii. Left Join v. Cross Join

ii. Outer Join iv. Right Join vi. Self-Join

b. Write the Syntax for Transforming Data and Example. **06**

i. CEILING() ii. Floor( ) iii. Case Conversion Function

iv. Date( ) v. Substring( ) vi. Power( )

c. For the following relation schema: **08**

employee(employee-name, street, city)

works(employee-name, company-name, salary)

company(company-name, city)

manages(employee-name, manager-name)

Give an expression in SQL for each of the following queries:

i. Find the names, street address, and cities of residence for all employees who work for 'First Bank Corporation' and earn more than $10,000.

ii. Find the names of all employees in the database who live in the same cities as the companies for which they work.

iii. Find the names of all employees in the database who live in the same cities and on the same streets as do their managers.

iv. Find the names of all employees in the database who do not work for 'First Bank Corporation'. Assume that all people work for exactly one company.

7. a. Illustrate ROWID function and ROWNUM Function. **04**

b. Explain How Index increase performance of SQL. Discuss Types of Index with Example. **08**

c. Write Syntax for following Command with Example. **08**

i. Creating View

ii. Views Multiple Tables

iii. Creating Index

iv. Create User

OR

8. a. Exemplify Vertical, Horizontal and Group view. Illustrate use of cascading & restrict option. List out limitations on view. **10**

b. List out Granting and Revoking Privileges. How Grant and Revoke execute with GRANT OPTION Clause? **10**